Year 2 Pure Mathematics

Chapter 2

Functions and graphs

Practice Booklet

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2.1 The modulus function

Practice Book, printed page 6.

1. $f(x) = |x^2 + 5x|$. Write down the value of:

a. $f(-3)$

b. $f(2)$

c. $f(-1)$

Hint

The modulus of a number a is its non-negative numerical value. It is written as $|a|$.
In general:

- $|f(x)| = f(x)$ when $f(x) \geq 0$
- $|f(x)| = -f(x)$ when $f(x) < 0$.

2. $g(x) = |2 - 3x| + 1$. Write down the value of:

a. $g(1)$

b. $g(10)$

c. $g(-1)$

3. Sketch each graph and mark the coordinates of any points of intersection with the coordinate axes.

a. $y = |x + 1|$

b. $y = |2x + 1|$

c. $y = |3x - 4|$

d. $y = |3 - 2x|$

e. $y = \left|\frac{1}{2}x + 3\right|$

f. $y = -|3x + 1|$

Hint

The graph of $y = |ax+b|$ is the graph of $y = ax+b$ with the section of the graph below the x-axis reflected in the x-axis.

4.

i. For each pair of functions, sketch $y = f(x)$ and $y = g(x)$ on the same set of axes.

ii. Hence solve $f(x) = g(x)$.

a. $f(x) = |2x - 3|$, $g(x) = 7$
b. $f(x) = |2 - 5x|$, $g(x) = 3x + 5$
c. $f(x) = |3x + 2|$, $g(x) = \frac{x}{2} + 5$

Hint

You can solve $f(x) = g(x)$ algebraically by solving two equations. For example, in part a:

$$2x - 3 = 7 \qquad (1)$$

$$-(2x - 3) = 7 \qquad (2)$$

5. Solve the inequality $|4x - 5| > 2x$.
Write your answer in set notation.

Hint

Solve $|4x - 5| = 2x$, then use sketches of $y = |4x - 5|$ and $y = 2x$ to work out which values of x satisfy the inequality.
← Year 1, Section 3.4

6. E
a. Sketch the graph with equation $y = |4x + 3|$, marking the locations of any points where the graph cuts or meets the coordinate axes. (2 marks)
b. Find the values of x which satisfy $|4x + 3| \leq 7$. (2 marks)

7. E
a. $f(x) = |5 - 3x|$. Sketch the graph with equation $y = f(x)$, stating the coordinates of any points where the graph cuts or meets the coordinate axes. (2 marks)
b. Find the values of x for which $f(x) = x$. (4 marks)

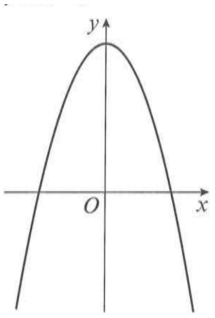
8. E/P Solve the inequality $|2x - 6| > 5 - x$. (4 marks)

9. E/P
a. Sketch the graph with equation $y = -|3 - x|$. (2 marks)
b. Find the values of x which satisfy $-|3 - x| = -\frac{x + 5}{3}$ (5 marks)

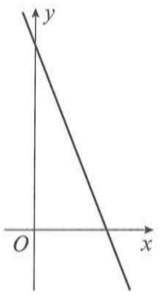
2.2 Functions and mappings

Practice Book, printed page 7.

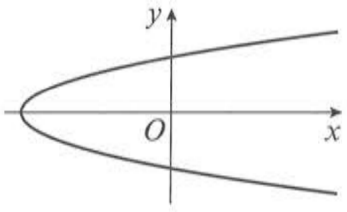
1. State whether each mapping from the set of x-values to the set of y-values is one-to-one, many-to-one or one-to-many.
a. $y = 5 - x^2$



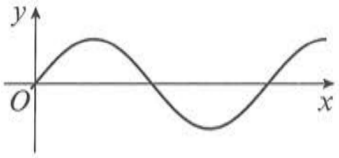
b. $y = 10 - 3x$



c. $y^2 = x + 9$



d. $y = \sin x$



Hint

A **mapping** transforms one set of numbers, the **domain**, onto another set of numbers, the **range**.

2. State a possible domain which would make each mapping a function and write it using function notation.

a. $y = \sqrt{x+2}$

b. $y = \dfrac{1}{x+1}$

c. $y = \tan x$

Hint

A mapping is a **function** if every element in the domain maps to one distinct element in the range. This means functions are either one-to-one or many-to-one.
It is possible to convert some mappings into functions by restricting the domain.

3. For each function:

i. sketch the graph of $y = \mathrm{f}(x)$

ii. state the range of $\mathrm{f}(x)$

iii. state whether the function is one-to-one or many-to-one.

a. $\mathrm{f}(x) = (x+2)^2$, $x \in \mathbb{R}$, $x \geq 0$

b. $\mathrm{f}(x) = 3\sin x$, $x \in \mathbb{R}$, $0 \leq x \leq 180°$

c. $\mathrm{f}(x) = 2e^x$, $x \in \mathbb{R}$, $x \geq 0$

d. $\mathrm{f}(x) = \sqrt{x-3}$, $x \in \mathbb{R}$, $x \geq 3$

Hint

You should only sketch the graph of the function for values of x in its **domain**.
The **range** will be the set of y-values that are mapped to by the function.

4. The function $\mathrm{f}(x)$ is defined by

$$\mathrm{f}(x) = \begin{cases} 5 - \frac{1}{2}x, & x \in \mathbb{R},\ x < 2 \\ (x-4)^2 + 2, & x \in \mathbb{R},\ x \geq 2 \end{cases}$$

a. Sketch the graph of $y = \mathrm{f}(x)$.
b. Explain why $\mathrm{f}(x)$ is a function and state the value of $\mathrm{f}(2)$.
c. State the range of $\mathrm{f}(x)$.
d. Solve $\mathrm{f}(x) = 11$.

Hint

This is an example of a **piecewise-defined function**. It consists of two parts, one linear for $x < 2$ and one quadratic for $x \geq 2$. The function is not continuous as it jumps at $x = 2$ but it is still defined for all real values of x.

5. E The function f is defined by $\mathrm{f} : x \mapsto x^2 - 3x + 4$, $x \in \mathbb{R}$, $0 \leq x < 4$.
a. Find the range of $\mathrm{f}(x)$. (3 marks)
b. State whether the function is one-to-one or many-to-one. (1 mark)

6. E/P The function $\mathrm{f}(x)$ is defined by

$$\mathrm{f}(x) = \begin{cases} e^x + 5, & x \in \mathbb{R},\ x \leq 0 \\ x^2 - 8x + 6, & x \in \mathbb{R},\ x \geq 0 \end{cases}$$

a. Sketch the graph of $y = \mathrm{f}(x)$. (4 marks)
b. State the range of $\mathrm{f}(x)$. (1 mark)

7. E/P The function g defined by $\mathrm{g}(x) = x^2 - 5x + 8$ has domain $x \in \mathbb{R}$, $x \geq a$. Given that g is a one-to-one function, find the smallest possible value of the constant a. (6 marks)

8. E/P The function $\mathrm{h}(x)$ is defined by

$$\mathrm{h}(x) = \begin{cases} 2 + x, & x \in \mathbb{R},\ x \leq 4 \\ 10x - x^2 - 17, & x \in \mathbb{R},\ x > 4 \end{cases}$$

a. Sketch the graph of $y = \mathrm{h}(x)$. (4 marks)
b. State the value of $\mathrm{h}(4)$. (1 mark)
c. State the range of $\mathrm{h}(x)$. (1 mark)
d. Find the exact values of x for which $\mathrm{h}(x) = 5$. (4 marks)

2.3 Composite functions

Practice Book, printed page 8.

1. Given $\mathrm{f}(x) = x^3$, $\mathrm{g}(x) = x - 2$ and $\mathrm{h}(x) = \frac{x}{2}$, find:
a. $\mathrm{fg}(4)$
b. $\mathrm{gf}(2)$
c. $\mathrm{g}^2(-1)$
d. $\mathrm{gh}(6)$
e. $\mathrm{fgh}(10)$

Hint

$\mathrm{fg}(x)$ means apply g first, then apply f, so $\mathrm{fg}(x) = \mathrm{f}(\mathrm{g}(x))$.

2. Given the functions $\mathrm{p}(x) = 3x - 1$, $\mathrm{q}(x) = 4 - x^2$ and $\mathrm{r}(x) = \frac{1}{x}$, find expressions for:

a. $\mathrm{pq}(x)$
b. $\mathrm{qr}(x)$
c. $\mathrm{p}^2(x)$
d. $\mathrm{rp}(x)$
e. $\mathrm{pqr}(x)$

Hint

In part a, $\mathrm{pq}(x) = 3(4 - x^2) - 1$.
Simplify your answers as much as possible.

3. The functions f and g are defined by $\mathrm{f} : x \mapsto |4x - 5|$ and $\mathrm{g} : x \mapsto \dfrac{x-1}{2}$ respectively.

a. Find $\mathrm{fg}(-5)$.
b. Solve $\mathrm{fg}(x) = 2x$.

Hint

For part b, sketch $y = \mathrm{fg}(x)$ and $y = 2x$ on the same set of axes. The number of points of intersection will be the number of solutions to the equation $\mathrm{fg}(x) = 2x$.

4. E/P $\mathrm{f}(x) = e^x$, $x \in \mathbb{R}$ and $\mathrm{g}(x) = 2\ln x$, $x \in \mathbb{R}$, $x > 0$.

a. Find an expression for $\mathrm{gf}(x)$, simplifying your answer. (2 marks)
b. Show that there is only one real value of x for which $\mathrm{gf}(x) = \mathrm{fg}(x)$. (3 marks)

5. E/P $\mathrm{f}(x) = 4x^2 - 8x$, $x \in \mathbb{R}$ and $\mathrm{g}(x) = k - x$, $x \in \mathbb{R}$ where k is a non-zero constant.

a. Find the range of $\mathrm{f}(x)$. (3 marks)
b. Find an expression for $\mathrm{gf}(x)$ in terms of x and k. (2 marks)
c. Given that the range of gf is $\mathrm{gf}(x) < 6$, find the set of possible values of k. (2 marks)

6. E/P The function f is defined for all real values of x as $\mathrm{f}(x) = 8 - (x - 2)^2$.

a. State the range of $\mathrm{f}(x)$. (1 mark)
b. Find the value of $\mathrm{ff}(-2)$. (3 marks)

7. E/P Given that $\mathrm{f}(x) = 2x^2$, $x \in \mathbb{R}$ and $\mathrm{g}(x) = \dfrac{1}{x-1}$, $x \in \mathbb{R}$, $x \neq 1$,

a. state the range of $\mathrm{f}(x)$ (1 mark)
b. find an expression for $\mathrm{fg}(x)$ (2 marks)
c. solve the equation $\mathrm{fg}(x) = 18$. (4 marks)

8. E/P The functions p and q are defined by $\mathrm{p} : x \mapsto 3^x$, $x \in \mathbb{R}$ and $\mathrm{q} : x \mapsto x - 2$, $x \in \mathbb{R}$ respectively.

a. Find an expression for $\mathrm{pq}(x)$. (2 marks)
b. Find an expression for $\mathrm{qp}(x)$. (2 marks)
c. Solve $\mathrm{pq}(x) = \mathrm{qp}(x)$, giving your answer in the form $\dfrac{\ln a}{\ln b}$ (3 marks)

2.4 Inverse functions

Practice Book, printed page 9.

1. For each function $\mathrm{f}(x)$,

i. determine the equation of the inverse function $\mathrm{f}^{-1}(x)$
ii. sketch the graphs of $y = x$, $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$ on the same set of axes.

a. $\mathrm{f} : x \mapsto 2x - 5$, $x \in \mathbb{R}$
b. $\mathrm{f} : x \mapsto \dfrac{x-3}{2}$, $x \in \mathbb{R}$
c. $\mathrm{f} : x \mapsto x^3 - 1$, $x \in \mathbb{R}$
d. $\mathrm{f} : x \mapsto 1 - 3x$, $x \in \mathbb{R}$

Hint

The **inverse** of a function maps the elements in the range of the function back onto the original elements in its domain. The graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$ are reflections in the line $y = x$.

2. For each function $\mathrm{g}(x)$,

i. determine the equation of the inverse function $\mathrm{g}^{-1}(x)$

ii. state the domain and range of $\mathrm{g}^{-1}(x)$

iii. sketch the graphs of $y = \mathrm{g}(x)$ and $y = \mathrm{g}^{-1}(x)$ on the same set of axes

iv. solve the equation $\mathrm{g}(x) = \mathrm{g}^{-1}(x)$.

a. $\mathrm{g}(x) = 2x - 3$, $x \in \mathbb{R}$, $x \geq 0$

b. $\mathrm{g}(x) = \sqrt{x - 2}$, $x \in \mathbb{R}$, $x > 2$

c. $\mathrm{g}(x) = \dfrac{4}{x - 3}$, $x \in \mathbb{R}$, $x > 3$

d. $\mathrm{g}(x) = (x - 1)^2 - 5$, $x \in \mathbb{R}$, $x \geq 1$

Hint

The domain of $\mathrm{g}(x)$ is the range of $\mathrm{g}^{-1}(x)$ and the range of $\mathrm{g}(x)$ is the domain of $\mathrm{g}^{-1}(x)$.

3. $\mathrm{f}(x) = \dfrac{2}{x}$, $x \in \mathbb{R}$, $x \neq 0$

a. Show that f is self-inverse.

b. Write down an expression for $\mathrm{ff}(x)$.

Hint

A function $\mathrm{f}(x)$ is **self-inverse** if $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$.

4. The function g is defined by

$$\mathrm{g} : x \mapsto x^2 - 4x, \ x \in \mathbb{R}, \ 0 \leq x \leq 6$$

Explain why the function g does not have an inverse.

Hint

Sketch the function to check whether it is one-to-one for the given domain.

5. E/P The function $\mathrm{f}(x)$ is defined by $\mathrm{f}(x) = \dfrac{3x - 5}{x - 3}$, $x \in \mathbb{R}$, $x \neq 3$.

a. Show that $\mathrm{ff}(x) = x$. (3 marks)

b. Hence write down $\mathrm{f}^{-1}(x)$. (1 mark)

6. E/P The function f is defined by $\mathrm{f} : x \mapsto \ln(3x - 1)$, $x \in \mathbb{R}$, $x > \dfrac{1}{3}$

a. Find $\mathrm{f}^{-1}(x)$ and state its domain and range. (4 marks)

b. Sketch the graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$ on the same set of axes, stating the exact coordinates of the points where the graphs cross the axes. (4 marks)

7. E/P The function $\mathrm{g}(x)$ is defined by $\mathrm{g}(x) = x^2 - 6x + 8$, $x \in \mathbb{R}$, $x > 3$.

a. Find $\mathrm{g}^{-1}(x)$ and state its domain and range. (6 marks)

b. Find the exact value of the solution to the equation $\mathrm{g}(x) = \mathrm{g}^{-1}(x)$. (3 marks)

8. E/P The function f is defined by $\mathrm{f}(x) = \dfrac{2x + 1}{x - 3}$, $x \in \mathbb{R}$, $x > 3$.

a. Find $\mathrm{f}^{-1}(x)$ and state its domain and range. (4 marks)

b. Solve the equation $\mathrm{f}^{-1}(x) = 8$. (2 marks)

2.5 $y = |f(x)|$ and $y = f(|x|)$

Practice Book, printed page 10.

1. For each function $f(x)$, sketch the graphs of $y = f(x)$ and $y = |f(x)|$.

a. $f(x) = x^2 - x - 12$, $x \in \mathbb{R}$

b. $f(x) = \cos x$, $0 \leq x \leq 360°$

Hint

To sketch the graph of $y = |f(x)|$:

- Sketch the graph of $y = f(x)$.
- Reflect any parts of the graph below the x-axis, (where $f(x) < 0$) in the x-axis.
- Delete the parts of the graph below the x-axis.

2. For each function $f(x)$, sketch the graphs of $y = f(x)$ and $y = f(|x|)$.

a. $f(x) = 6 + 5x - x^2$, $x \in \mathbb{R}$

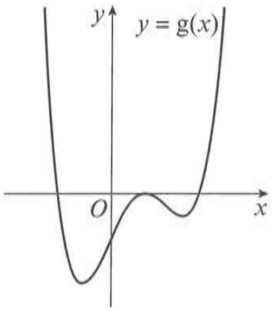
b. $f(x) = 2\sin x$, $-360° \leq x \leq 360°$

Hint

To sketch the graph of $y = f(|x|)$:

- Sketch the graph of $y = f(x)$ for $x \geq 0$.
- Reflect this in the y-axis.

3. The diagram shows a sketch of part of the curve with equation $y = g(x)$, $x \in \mathbb{R}$.

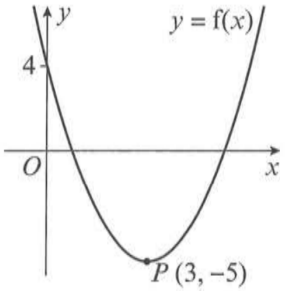


On separate diagrams, sketch the curves with equations:

a. $y = |g(x)|$

b. $y = g(|x|)$

4. E The diagram shows a sketch of the curve with the equation $y = \mathrm{f}(x)$, $x \in \mathbb{R}$.
The curve has a turning point at $P(3, -5)$ and also passes through the point $(0, 4)$.



a. Write down the coordinates of the point to which P is transformed on the curve with equation $y = |\mathrm{f}(x)|$. (2 marks)

b. Sketch the curve with equation $y = \mathrm{f}(|x|)$. On your sketch, show the coordinates of all turning points and the coordinates of the point where the curve cuts the y-axis. (3 marks)

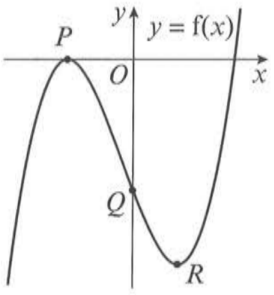
5. E Sketch the graphs of:

a. $y = |\ln x|$ (2 marks)

b. $y = \ln(|x|)$ (3 marks)

In each case, state the coordinates of any points of intersection with the axes.

6. E The diagram shows a sketch of the curve with equation $y = \mathrm{f}(x)$. The curve passes through the points $P(-4, 0)$, $Q(0, -10)$ and $R(3, -15)$.



On separate diagrams, sketch the graphs with equations:

a. $y = |\mathrm{f}(x)|$ (3 marks)

b. $y = \mathrm{f}(|x|)$ (3 marks)

On each sketch, where possible, show the coordinates of the points corresponding to P, Q and R.

7. E/P The function g is defined by $\mathrm{g} : x \mapsto \dfrac{3}{x-2}$, $x \in \mathbb{R}$, $x \neq 2$.

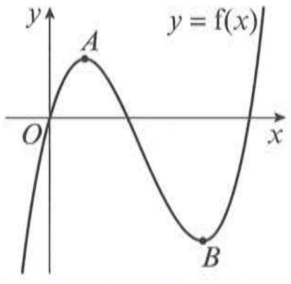
a. Sketch the graph of $y = |\mathrm{g}(x)|$. Indicate clearly the equation of the vertical asymptote and the coordinates of the point at which the graph crosses the y-axis. (3 marks)

b. Find the values of x for which $\left|\dfrac{3}{x-2}\right| = 4$. (3 marks)

2.6 Combining transformations

Practice Book, printed page 12.

1. The diagram shows a sketch of the graph of $y = \mathrm{f}(x)$.
The curve passes through the origin O, the point $A(2, 8)$ and the point $B(8, -16)$.

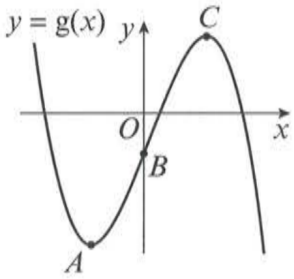


Sketch the graph of $y = \mathrm{f}(x + 3) + 4$ and find the coordinates of the images of the points A and B.

Hint

$y = \mathrm{f}(x) + a$ is a translation of the graph of $y = \mathrm{f}(x)$ by the vector $\begin{pmatrix} 0 \\ a \end{pmatrix}$
$y = \mathrm{f}(x + a)$ is a translation of the graph of $y = \mathrm{f}(x)$ by the vector $\begin{pmatrix} -a \\ 0 \end{pmatrix}$
← Year 1, Section 4.5

2. The diagram shows a sketch of the graph of $y = \mathrm{g}(x)$.
The curve passes through the points $A(-2, -7)$, $B(0, -2)$ and $C(2, 4)$.



Sketch the graphs of:

a. $y = 3\mathrm{g}(x - 1)$

b. $y = \mathrm{g}\left(\frac{x}{2}\right) - 2$

Find the coordinates of the images of the points A, B and C.

Hint

$y = a\mathrm{g}(x)$ is a vertical stretch by scale factor a of the graph of $y = \mathrm{g}(x)$.
$y = \mathrm{g}(ax)$ is a horizontal stretch by scale factor $\frac{1}{a}$ of the graph of $y = \mathrm{g}(x)$.
In part a, apply the translation first and then the stretch. ← Year 1, Section 4.6

3. $\mathrm{f}(x) = e^x$. Sketch the graphs with equations:

a. $y = -2\mathrm{f}(x) - 1$

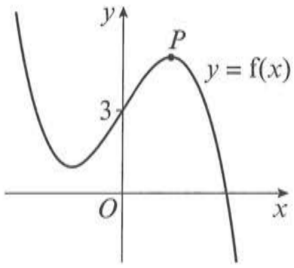
b. $y = -\mathrm{f}(|x|)$

Show, on each diagram, the point where the graph crosses the y-axis.
For part a, state the equation of the asymptote.

Hint

The graph of $y = -\mathrm{f}(x)$ is a reflection of the graph of $y = \mathrm{f}(x)$ in the x-axis.
These graphs can be considered as vertical stretches with scale factor -1.
← Year 1, Section 4.6

4. E/P The diagram shows a sketch of the graph of $y = \mathrm{f}(x)$.
The graph cuts the y-axis at the point $(0, 3)$ and the point $P(1, 5)$ is the local maximum turning point.



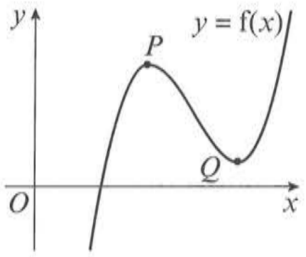
On separate sets of axes, sketch the graphs of:

a. $y = \mathrm{f}(-x) + 1$ (3 marks)

b. $y = \mathrm{f}(x + 2) + 3$ (3 marks)

On each sketch, show the coordinates of the point at which your graph intersects the y-axis, if known, and the coordinates of the point to which P is transformed.

5. E/P The diagram shows the graph of $y = \mathrm{f}(x)$.
The points $P(3, 5)$ and $Q(5, 1)$ are turning points on the graph.



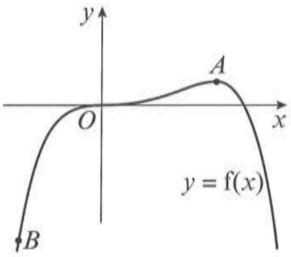
On separate diagrams, sketch the graphs of:

a. $y = -3\mathrm{f}(x) + 1$ (3 marks)

b. $y = \mathrm{f}(|x|) - 1$ (3 marks)

For each graph, indicate the coordinates of any turning points.

6. E/P The diagram shows a sketch of the curve with equation $y = \mathrm{f}(x)$.
The curve passes through the origin O and the points $A(9, 10)$ and $B(-9, -70)$.



On separate diagrams, sketch the graphs of:

a. $y = \dfrac{\mathrm{f}(x + 2)}{2}$ (3 marks)

b. $y = \mathrm{f}(2x) - 3$ (3 marks)

Indicate on each diagram the coordinates of the points corresponding to A and B.

7. E/P The function g is defined by $\mathrm{g} : x \mapsto (x + 1)^2 - 4$, $x \in \mathbb{R}$.

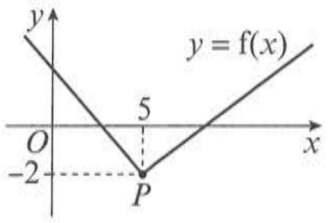
a. Draw a sketch of the graph of $y = \mathrm{g}(x)$, labelling the turning point and the x- and y-intercepts. (3 marks)

b. Write down the coordinates of the turning points on the graphs with equations:

i. $y = 2(|g(x)|) + 2$
ii. $y = -3g(x + 1)$
iii. $y = g(2x) + 1$

(6 marks)

8. E/P The diagram shows part of the graph of $y = f(x)$, $x \in \mathbb{R}$.
The graph consists of two line segments that meet at the point $P(5, -2)$, as shown in the diagram.



On separate diagrams, sketch the graphs of:

a. $y = 2f(x + 5)$ (3 marks)
b. $y = |f(-x)|$ (3 marks)

On each diagram, show the coordinates of the point corresponding to P.

2.7 Solving modulus problems

Practice Book, printed page 14.

1. The function f is defined as

$$f(x) = 2|x + 1| - 3, \ x \in \mathbb{R}$$

a. Sketch the graph of $y = f(x)$.
b. State the range of the function.
c. Solve the equation $f(x) = x + 4$.

Hint

Use transformations to sketch the graph.
Start with $y = |x|$, then apply a translation by the vector $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$, then a stretch with scale factor 2 in the y-direction, and then a translation by the vector $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$.

2. The function g is defined as

$$g(x) = 9 - 3|x - 2|, \ x \in \mathbb{R}$$

a. Sketch the graph of the function.
b. State the range of the function.
c. Give a reason why $g^{-1}(x)$ does not exist.
d. Solve the equation $g(x) \leq 3$, giving your answer in set notation.

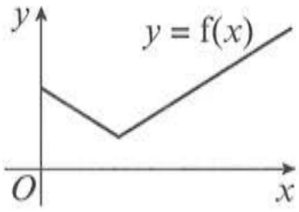
Hint

The part of the function inside the modulus signs must always be greater than or equal to 0.

3. For each function $f(x)$:
i. sketch the graph of $y = f(x)$
ii. state the range of the function.

a. $f(x) = 2|x| - 1$, $x \in \mathbb{R}$
b. $f(x) = 2 - 3|x + 1|$, $x \in \mathbb{R}$
c. $f(x) = 5 + \dfrac{|x - 2|}{2}$, $x \in \mathbb{R}$

4. E/P The diagram shows a sketch of part of the graph $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = 3|2 - x| + 4$, $x \geq 0$.



a. State the range of $\mathrm{f}(x)$. (1 mark)

b. Solve the equation $\mathrm{f}(x) = x + 20$. (3 marks)

c. Given that the equation $\mathrm{f}(x) = k$, where k is a constant, has two distinct roots, state the set of possible values for k. (2 marks)

5. E/P

a. Sketch the graph of $y = |x| - 5$. (3 marks)

b. Explain why $|x| - 5 \leq |x - 5|$ for all real values of x. (3 marks)

6. E/P

a. Sketch the graph with equation $y = |5x - 4|$, stating the coordinates of any points where the graph cuts or meets the axes. (2 marks)

Find the complete set of values of x for which:

b. $|5x - 4| > 3 - 2x$ (4 marks)

c. $|5x - 4| > \frac{8}{5} - 2x$ (2 marks)

7. E/P Given that p and q are positive constants,

a. on separate diagrams, sketch the graphs with equations:

i. $y = |3x - p|$

ii. $y = |3x - p| + q$

Show, on each sketch, the coordinates of any points at which the graph crosses or meets the axes. (4 marks)

Given that the equation $|3x - p| + q = 2x + 8$ has solutions $x = 0$ and $x = k$,

b. find k in terms of p. (4 marks)

Problem solving: Set A

Bronze

The functions f and g are defined by $f: x \mapsto 3 - 2x^3$, $x \in \mathbb{R}$ and $g: x \mapsto \frac{2}{x} - 5$, $x \in \mathbb{R}$, $x > 0$.

a. Find the inverse function $f^{-1}(x)$. (2 marks)

b. Show that the composite function gf is given by $gf: x \mapsto \frac{10x^3 - 13}{3 - 2x^3}$ (4 marks)

c. Solve $gf(x) = 1$. (3 marks)

Silver

The function f is defined by $f(x) = e^{2x} + 2$, $x \in \mathbb{R}$.

a. Find $f^{-1}(x)$. State the domain of this inverse function. (3 marks)

The function $g(x)$ is defined by $g(x) = \ln(x + 3)$, $x \in \mathbb{R}$, $x > -3$.

b. Find $fg(x)$ and state its range. (4 marks)

c. Solve $fg(x) = 66$. (3 marks)

Gold

The function f is defined by $f: x \mapsto \frac{x - 5}{x^2 + 2x - 3} - \frac{2}{x + 3} + 2$, $x \in \mathbb{R}$, $x > 1$.

a. Show that $f(x) = \frac{2x - 3}{x - 1}$, $x > 1$. (4 marks)

b. Find the range of $f(x)$. (2 marks)

c. Find $f^{-1}(x)$. State the domain of this inverse function. (3 marks)

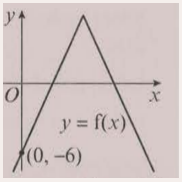
The function g is defined by $g: x \mapsto 3x^2 - 4$, $x \in \mathbb{R}$.

d. Solve $fg(x) = \frac{1}{5}$ (3 marks)

Problem solving: Set B

Bronze

The diagram shows a sketch of the graph $y = f(x)$, where $f(x) = k - 3|x - 4|$.
The graph cuts the y-axis at $(0, -6)$.



a. Find the value of k. (2 marks)

b. State the range of $f(x)$. (1 mark)

On separate diagrams, sketch the graphs of:

c.

i. $y = |f(x)|$

ii. $y = f(|x|)$

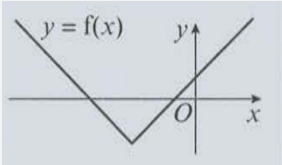
(4 marks)

d. Solve $f(x) = \frac{x}{2} + 1$. (5 marks)

Silver

The diagram shows a sketch of the graph $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2|x + 3| - 4$$



a. State the range of $\mathrm{f}(x)$. (1 mark)

On separate diagrams, sketch the graphs of:

b.

i. $y = |\mathrm{f}(x)|$

ii. $y = \mathrm{f}(|x|)$

(4 marks)

c. Find the set of values of x for which $\mathrm{f}(x) \leq 4 - 0.5x$. (5 marks)

d. Find the set of values of x for which $\mathrm{f}(x) = k$ has:

i. 0 solutions

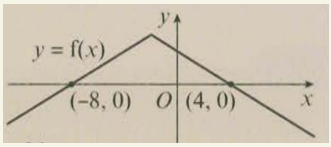
ii. 1 solution

iii. 2 solutions.

(3 marks)

Gold

$\mathrm{f}(x) = a - |x + b|$, $x \in \mathbb{R}$, where a and b are real constants.
The diagram shows the graph with equation $y = \mathrm{f}(x)$.
The graph cuts the x-axis at the points $(-8, 0)$ and $(4, 0)$.



a. Find the values of a and b. (3 marks)

b. Find the set of values of p for which the equation $\mathrm{f}(x) = px + 5$ has:

i. 0 solutions

ii. 1 solution

iii. 2 solutions.

(6 marks)

Now try this ⟶ Exam question bank Q26, Q32, Q55, Q63, Q66, Q79, Q82, Q85

Selected exam question bank questions

Question 26

The function f is defined by $\mathrm{f}(x) = 6x^3 - 13x^2 + 4$, $x \in \mathbb{R}$.

Given that $\mathrm{f}(2) = 0$, express $\mathrm{f}(x)$ as the product of three linear factors and hence sketch the graph of $y = |\mathrm{f}(x)|$. Show on your sketch the coordinates of each point at which the graph meets or cuts the coordinate axes. (7 marks)

Question 32

The function f is defined by $\mathrm{f}(x) = |2x - 7| + 3$.

a. Sketch the graph with equation $y = \mathrm{f}(x)$, stating the coordinates of any points where the graph cuts or meets the coordinate axes. (3 marks)

b. State the range of $\mathrm{f}(x)$. (1 mark)

c. Find the values of x which satisfy $\mathrm{f}(x) > x + 2$. Write your answer using set notation. (5 marks)

Question 55

a. Solve the equation $2|x + 4| - 2 = x + 3$. (4 marks)

b.

i. Sketch the graph of $y = |x| - 4$.

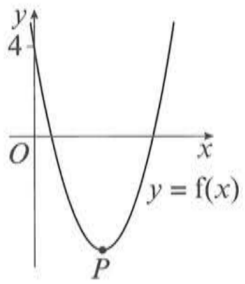
ii. Explain why $|x| - 4 \leq |x - 4|$ for all real values of x.

(6 marks)

Question 63

The diagram shows a sketch of the curve with equation $y = \mathrm{f}(x)$, $x \in \mathbb{R}$.

The curve passes through the point $(0, 4)$ and has a turning point at $P(3, -5)$.



a. Write down the coordinates of the point to which P is transformed on the curve with equation:

i. $2\mathrm{f}\left(\frac{x}{2}\right)$

ii. $-\mathrm{f}(-x)$

(4 marks)

b. Sketch the curve with equation $y = \mathrm{f}(|x|)$.
On your sketch show the coordinates of all turning points and the coordinates of the point at which the curve cuts the y-axis. (3 marks)

The curve with equation $y = \mathrm{f}(x)$ is a translation of the curve with equation $y = x^2$.

c. Find $\mathrm{f}(x)$. (2 marks)

d. Explain why the function f does not have an inverse. (1 mark)

Question 66

$\mathrm{f}(x) = \dfrac{2x-7}{x-3} + \dfrac{1}{(x-3)(x-2)}$, $x > 2$

a. Express $\mathrm{f}(x)$ as a single fraction in its simplest form. (5 marks)

b. Hence show that $\mathrm{f}(x) = 2 - \dfrac{1}{x-2}$, $x > 2$. (2 marks)

The curve $y = \dfrac{1}{x}$, $x > 0$, is mapped onto the curve $y = \mathrm{f}(x)$, using three successive transformations, T_1, T_2 and T_3.

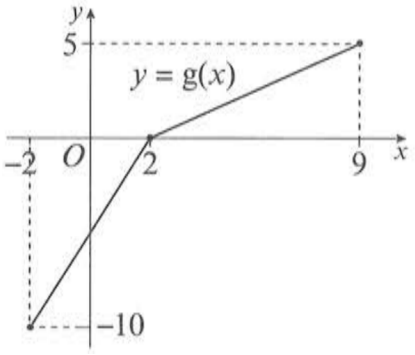
c. Describe fully T_1, T_2 and T_3. (4 marks)

Question 79

The function f is defined by $\mathrm{f}(x) = \dfrac{2-3x}{x-4}$, $x \in \mathbb{R}$, $x \neq 4$.

a. Find $\mathrm{f}^{-1}(x)$, and state the domain of this inverse function. (4 marks)

The diagram shows a sketch of the function $\mathrm{g}(x)$.
The function g has domain $-2 \leq x \leq 9$, and is linear from $(-2, -10)$ to $(2, 0)$ and from $(2, 0)$ to $(9, 5)$.



b. Write down the range of g. (1 mark)

c. Find $\mathrm{gg}(2)$. (2 marks)

d. Find $\mathrm{fg}(9)$. (2 marks)

e. On separate diagrams, sketch the graphs with equations:

i. $y = |\mathrm{g}(x)|$

ii. $y = \mathrm{g}^{-1}(x)$

(4 marks) Show on each sketch the coordinates of each point at which the graph meets or cuts the axes.

f. State the domain of the inverse function g^{-1}. (1 mark)

Question 82

The functions f and g are defined by $\mathrm{f}(x) = 3x + \ln 3$, $x \in \mathbb{R}$ and $\mathrm{g}(x) = e^x$, $x \in \mathbb{R}$.

a. Prove that the composite function gf is $\mathrm{gf}(x) = 3e^{3x}$, $x \in \mathbb{R}$. (4 marks)

b. Sketch the graph of $y = \mathrm{gf}(x)$, stating the coordinates of the point where the curve crosses the y-axis. (2 marks)

c. Write down the range of gf. (1 mark)

d. Find the value of x for which $\dfrac{\mathrm{d}}{\mathrm{d}x}(\mathrm{gf}(x)) = 4$, giving your answer to 3 significant figures. (4 marks)

Question 85

The function f is defined by $\mathrm{f}(x) = e^{-x} + 5$, $x \in \mathbb{R}$.

a. State the range of $\mathrm{f}(x)$. (2 marks)

b. Find $\mathrm{f}^{-1}(x)$. State the domain of this inverse function. (4 marks)

c. Sketch the graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$ on the set of same axes, stating the coordinates of the points where the graphs cross the axes. (4 marks)

d. Show that the x-coordinate of the point of intersection of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$ is the solution to the equation $e^x(x - 5) = 1$. (4 marks)

Answers

2.1 The modulus function

1.

a. 6

b. 14

c. 4

2.

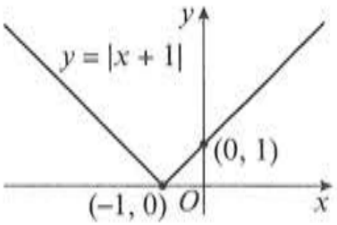
a. 2

b. 29

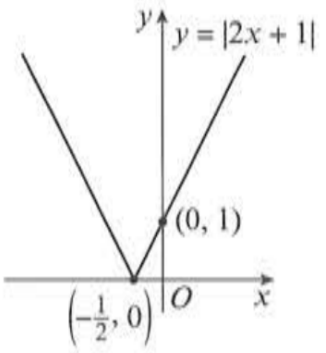
c. 6

3.

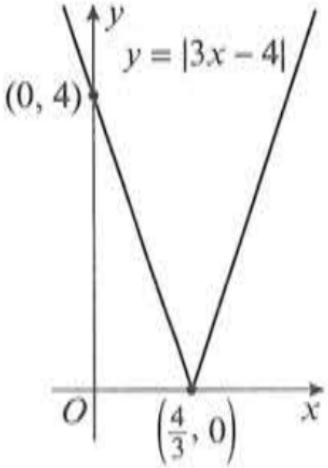
a.



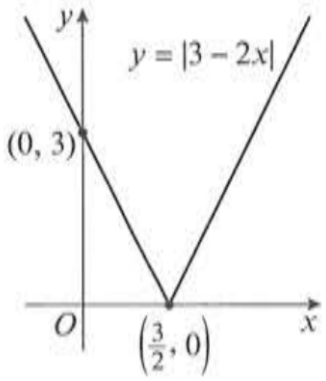
b.



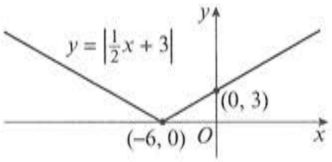
c.



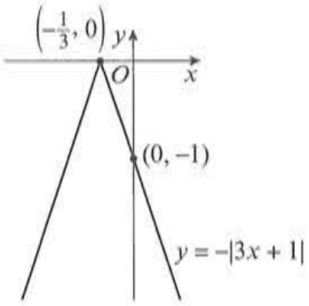
d.



e.



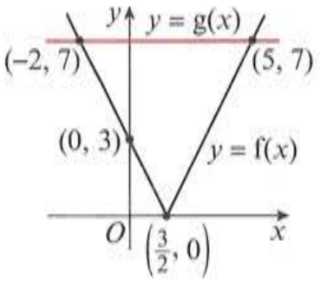
f.



4.

a.

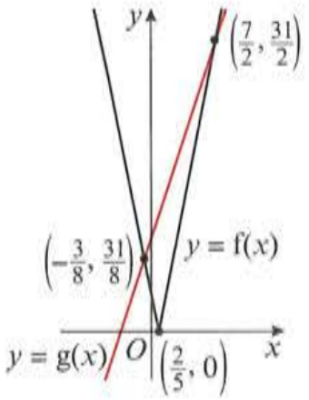
i.



ii. $x = -2, 5$

b.

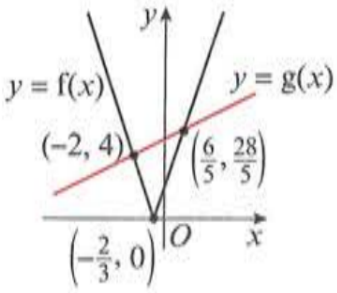
i.



ii. $x = -\frac{3}{8}, \frac{7}{2}$

c.

i.

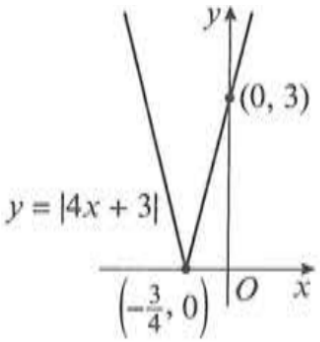


ii. $x = -2, \frac{6}{5}$

5. $\left\{x : x < \frac{5}{6}\right\} \cup \left\{x : x > \frac{5}{2}\right\}$

6.

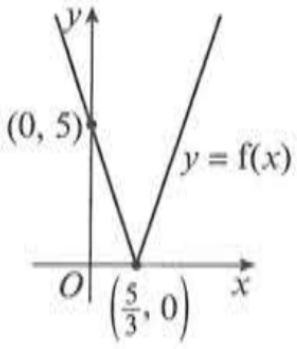
a.



b. $-\frac{5}{2} \leq x \leq 1$

7.

a.



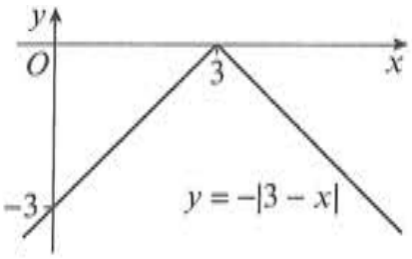
$(0, 5)$ and $\left(\frac{5}{3}, 0\right)$

b. $\frac{5}{4}$ and $\frac{5}{2}$

8. $x < 1$ or $x > \frac{11}{3}$

9.

a.



b. 1 and 7

2.2 Functions and mappings

1.

a. many-to-one

b. one-to-one

c. one-to-many

d. many-to-one

2.

a. $\mathrm{f} : x \mapsto \sqrt{x+2}$, $x \in \mathbb{R}$, $x \geq -2$

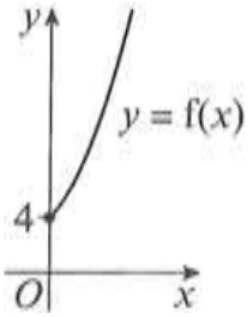
b. $\mathrm{f} : x \mapsto \frac{1}{x+1}$, $x \in \mathbb{R}$, $x \neq -1$

c. $\mathrm{f} : x \mapsto \tan x$, $x \in \mathbb{R}$, $-90° < x < 90°$

3.

a.

i.

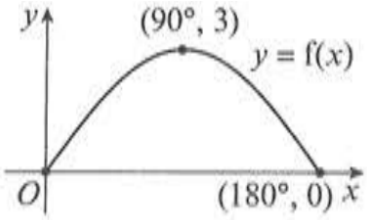


ii. $f(x) \geq 4$

iii. one-to-one

b.

i.

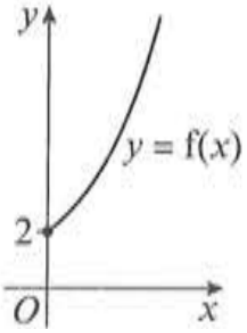


ii. $0 \leq f(x) \leq 3$

iii. many-to-one

c.

i.

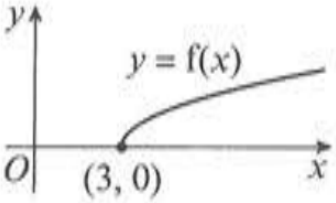


ii. $f(x) \geq 2$

iii. one-to-one

d.

i.

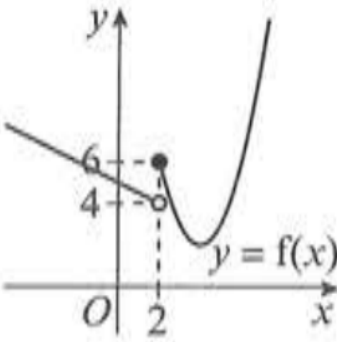


ii. $f(x) \geq 0$

iii. one-to-one

4.

a.



b. f(x) is a one-to-one mapping for all $x \in \mathbb{R}$, so it is a function. f$(2) = 6$

c. f$(x) \geq 2$

d. $x = -12$ or $x = 7$

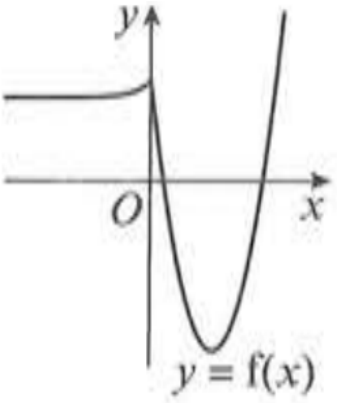
5.

a. $\frac{7}{4} \leq \text{f}(x) < 8$

b. many-to-one

6.

a.

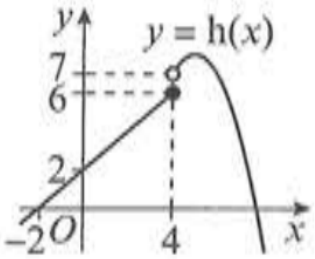


b. f$(x) \geq -10$

7. $\frac{5}{2}$

8.

a.



b. 6

c. h$(x) \leq 8$

d. 3 and $5 + \sqrt{3}$

2.3 Composite functions

1.

a. 8

b. 6
c. -5
d. 1
e. 27

2.

a. $11 - 3x^2$
b. $4 - \dfrac{1}{x^2}$
c. $9x - 4$
d. $\dfrac{1}{3x - 1}$
e. $11 - \dfrac{3}{x^2}$

3.

a. 17
b. $x = \dfrac{7}{4}$

4.

a. $\mathrm{gf}(x) = 2x$
b. $\mathrm{fg}(x) = x^2$. When $\mathrm{gf}(x) = \mathrm{fg}(x)$, $2x = x^2$, so $x = 0$ or $x = 2$
As $x = 0$ is not in the domain, $x = 2$.

5.

a. $\mathrm{f}(x) \geq -4$
b. $\mathrm{gf}(x) = k + 8x - 4x^2$
c. $\{k : k \in \mathbb{R}, k < 2\}$

6.

a. $\mathrm{f}(x) \leq 8$
b. -92

7.

a. $\mathrm{f}(x) \geq 0$
b. $2\left(\dfrac{1}{x - 1}\right)^2$
c. $x = \dfrac{2}{3}, \dfrac{4}{3}$

8.

a. 3^{x-2}
b. $3^x - 2$
c. $\dfrac{\ln \frac{9}{4}}{\ln 3}$

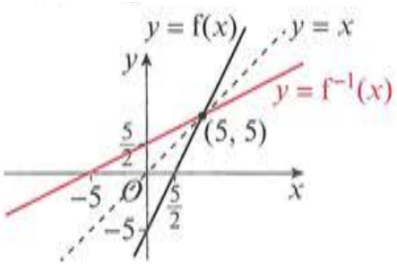
2.4 Inverse functions

1.

a.

i. $\mathrm{f}^{-1}(x) = \dfrac{x + 5}{2}$

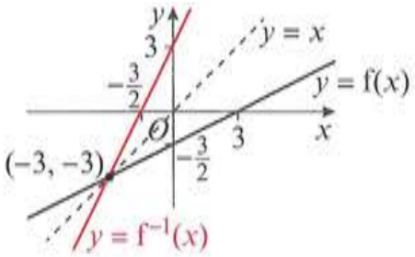
ii.



b.

i. $\mathrm{f}^{-1}(x) = 2x + 3$

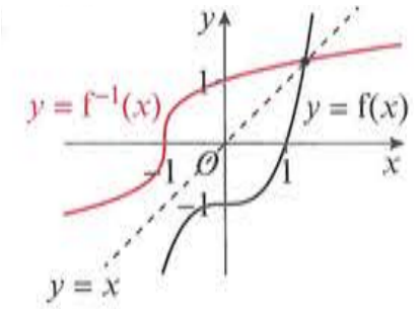
ii.



c.

i. $\mathrm{f}^{-1}(x) = \sqrt[3]{x+1}$

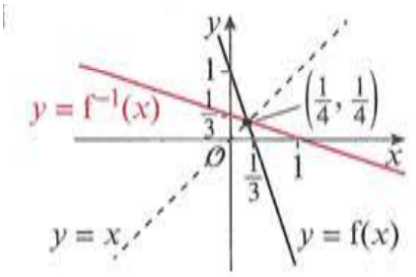
ii.



d.

i. $\mathrm{f}^{-1}(x) = \dfrac{1-x}{3}$

ii.



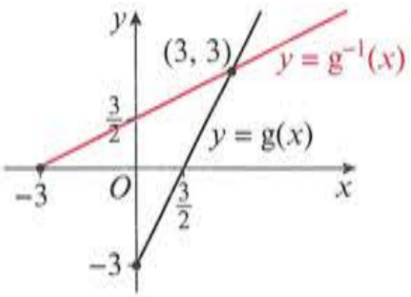
2.

a.

i. $\mathrm{g}^{-1}(x) = \dfrac{x+3}{2}$

ii. domain $x \in \mathbb{R}$, $x \geq -3$; range $\mathrm{g}^{-1}(x) \geq 0$

iii.



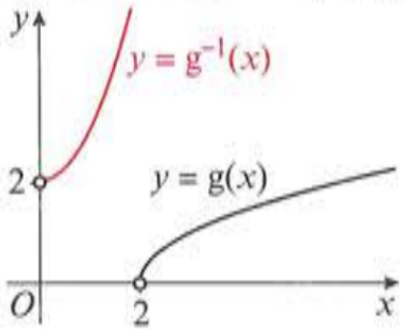
iv. $x = 3$

b.

i. $g^{-1}(x) = x^2 + 2$

ii. domain $x \in \mathbb{R}$, $x > 0$; range $g^{-1}(x) > 2$

iii.



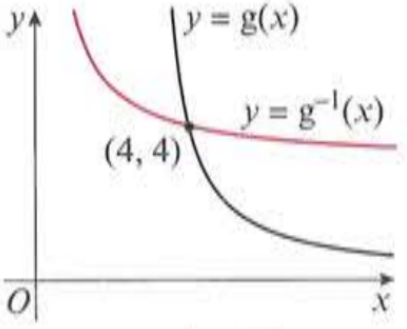
iv. no solutions

c.

i. $g^{-1}(x) = \dfrac{4 + 3x}{x}$

ii. domain $x \in \mathbb{R}$, $x > 0$; range $g^{-1}(x) > 3$

iii.



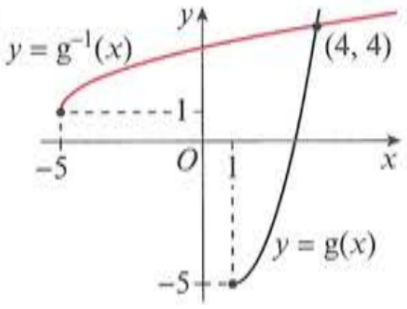
iv. $x = 4$

d.

i. $g^{-1}(x) = 1 + \sqrt{x + 5}$

ii. domain $x \in \mathbb{R}$, $x \geq -5$; range $g^{-1}(x) \geq 1$

iii.



iv. $x = 4$

3.

a. Let $f(x) = y = \dfrac{2}{x}$, then $xy = 2$ and $x = \dfrac{2}{y}$ so $f^{-1}(x) = \dfrac{2}{x} = f(x)$ and f is self-inverse.

b. x

4. $g(x)$ is not one-to-one in the interval $0 \leq x \leq 6$

5.

a. $\mathrm{f}\left(\dfrac{3x-5}{x-3}\right) = \dfrac{3\left(\frac{3x-5}{x-3}\right)-5}{\left(\frac{3x-5}{x-3}\right)-3} = \dfrac{3(3x-5)-5(x-3)}{3x-5-3(x-3)}$

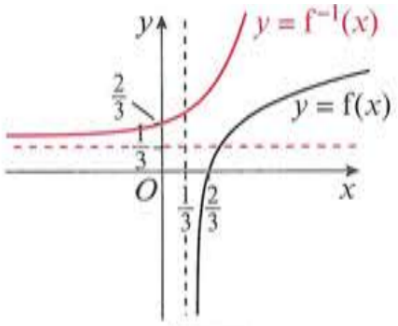
$= \dfrac{9x-15-5x+15}{3x-5-3x+9} = \dfrac{4x}{4} = x$

b. $\mathrm{f}^{-1}(x) = \dfrac{3x-5}{x-3}$, $x \in \mathbb{R}$, $x \neq 3$

6.

a. $\mathrm{f}^{-1}(x) = \dfrac{e^x+1}{3}$, $x \in \mathbb{R}$; $\mathrm{f}^{-1}(x) > \dfrac{1}{3}$

b.



7.

a. $\mathrm{g}^{-1}(x) = 3 + \sqrt{x+1}$, $x \in \mathbb{R}$, $x > -1$; $\mathrm{g}^{-1}(x) > 3$

b. $x = \dfrac{7+\sqrt{17}}{2}$

8.

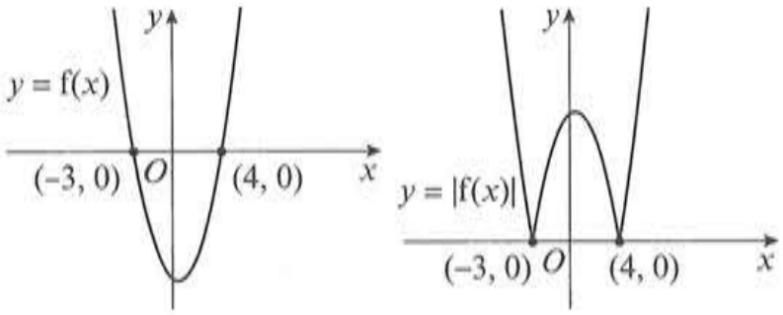
a. $\mathrm{f}^{-1}(x) = \dfrac{3x+1}{x-2}$, $x \in \mathbb{R}$, $x > 2$; $\mathrm{f}^{-1}(x) > 3$

b. $x = \dfrac{17}{5}$

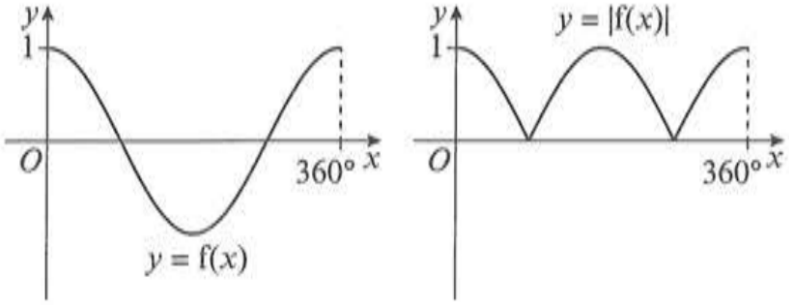
2.5 $y = |\mathrm{f}(x)|$ and $y = \mathrm{f}(|x|)$

1.

a.

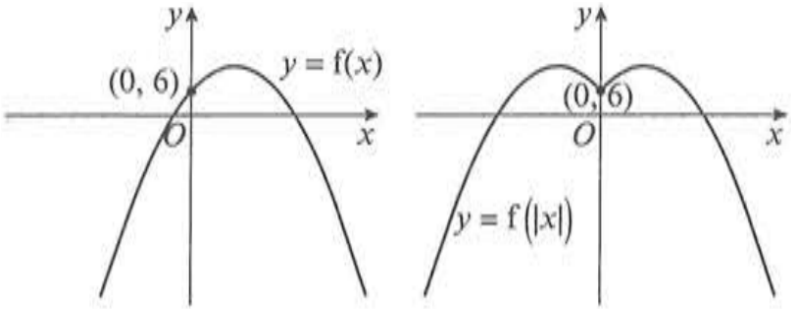


b.

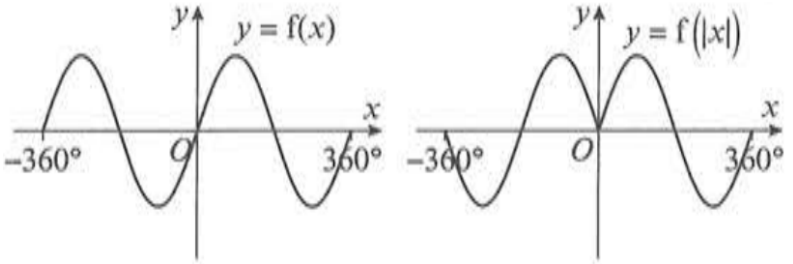


2.

a.

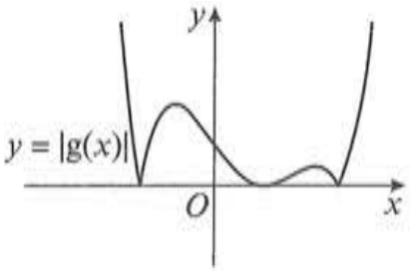


b.

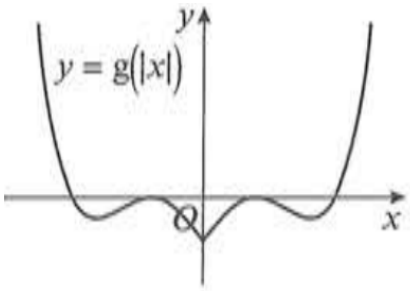


3.

a.



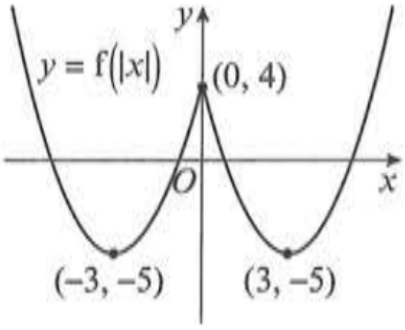
b.



4.

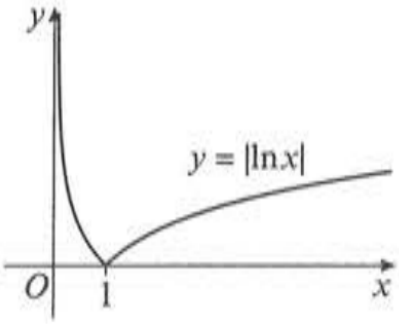
a. $(3, 5)$

b.

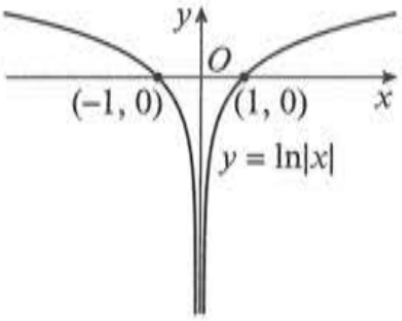


5.

a.

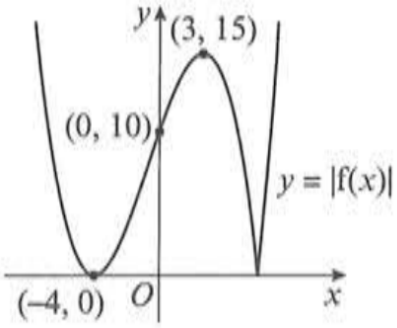


b.

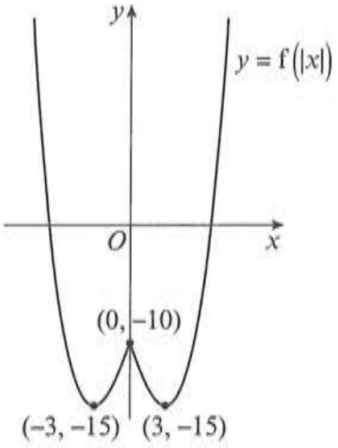


6.

a.

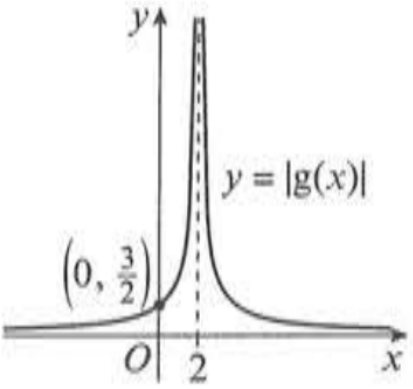


b.



7.

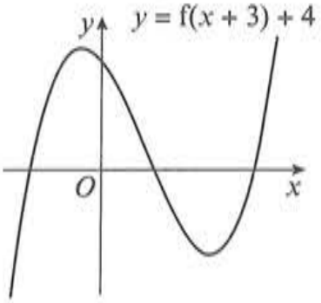
a.



b. $\frac{5}{4}$ and $\frac{11}{4}$

2.6 Combining transformations

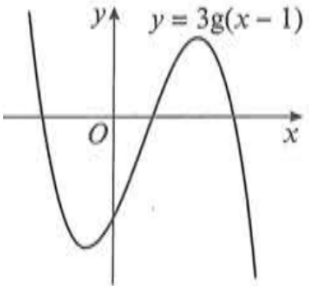
1.



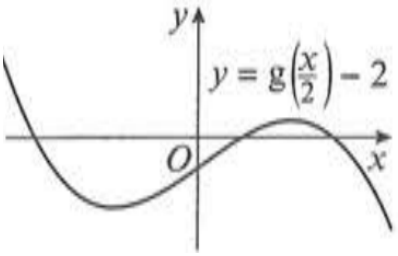
$A'(-1, 12)$, $B'(5, -12)$

2.

a. $A'(-1, -21)$, $B'(1, -6)$, $C'(3, 12)$

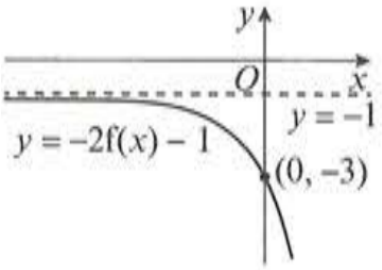


b. $A'(-4, -9)$, $B'(0, -4)$, $C'(4, 2)$

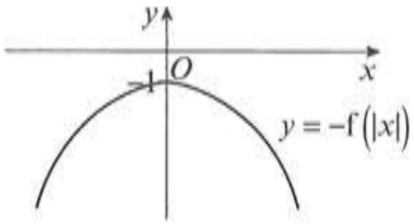


3.

a.

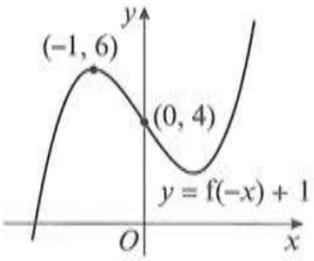

y
O
x
y = −1
y = −2f(x) − 1
(0, −3)

b.

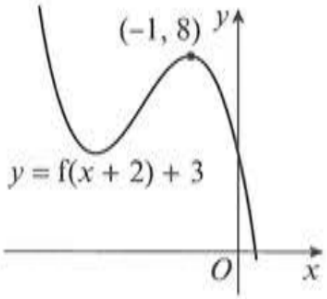

y
O
x
−1
y = −f(|x|)

4.

a.

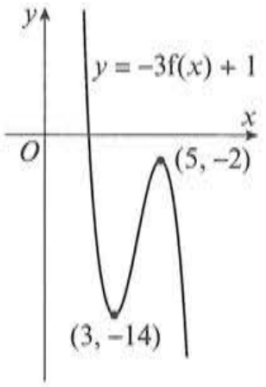

y
(−1, 6)
(0, 4)
y = f(−x) + 1
O
x

b.

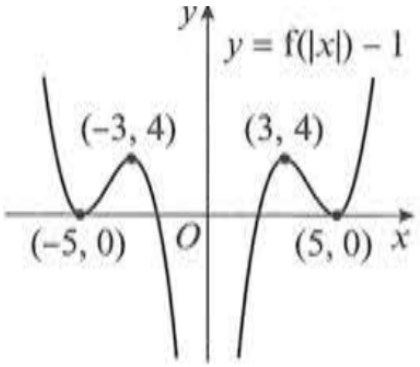

(−1, 8)
y
y = f(x + 2) + 3
O
x

5.

a.

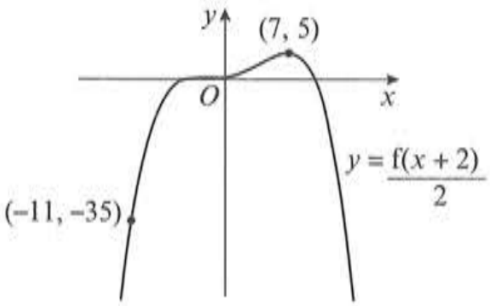

y
y = −3f(x) + 1
x
O
(5, −2)
(3, −14)

b.

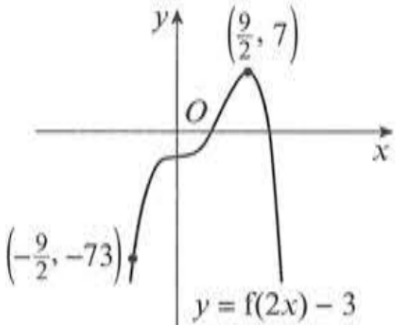


6.

a.

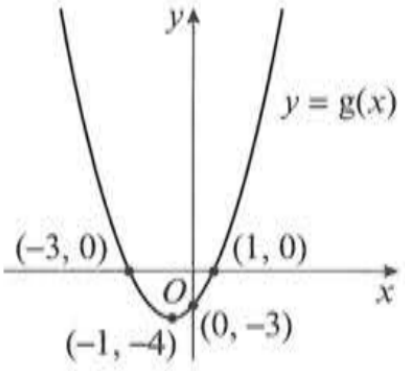


b.



7.

a.



b.

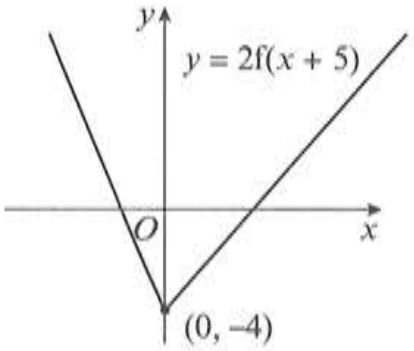
i. $(-1, 10)$

ii. $(-2, 12)$

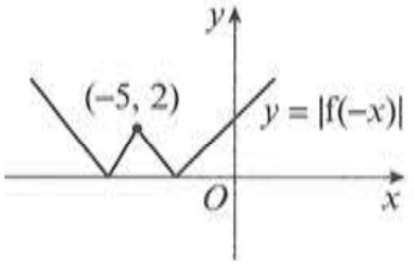
iii. $\left(-\frac{1}{2}, -3\right)$

8.

a.



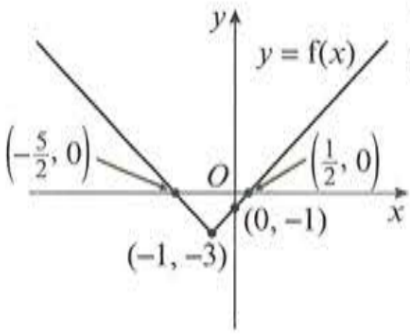
b.



2.7 Solving modulus problems

1.

a.

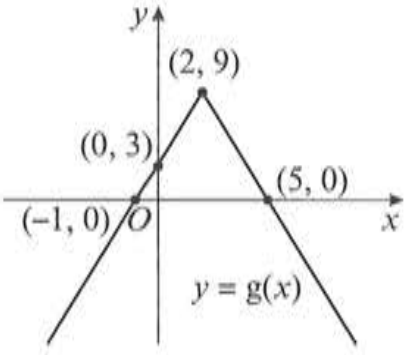


b. $f(x) \geq -3$

c. $x = -3$ or $x = 5$

2.

a.



b. $g(x) \leq 9$

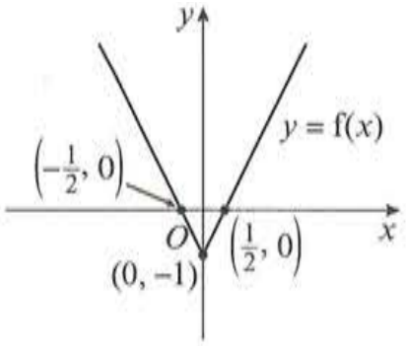
c. $g(x)$ is not a one-to-one function

d. $\{x : x \leq 0\} \cup \{x : x \geq 4\}$

3.

a.

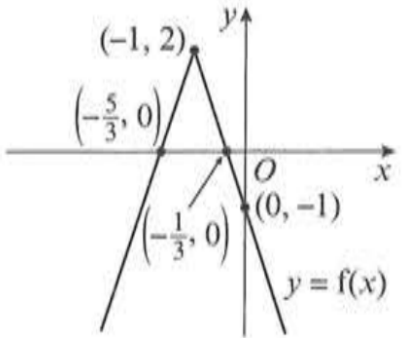
i.



ii. $\mathrm{f}(x) \geq -1$

b.

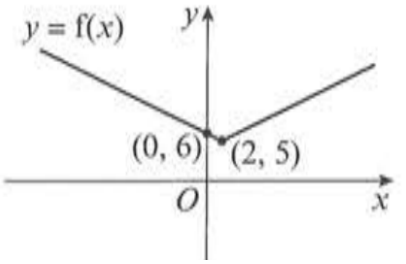
i.



ii. $\mathrm{f}(x) \leq 2$

c.

i.



ii. $\mathrm{f}(x) \geq 5$

4.

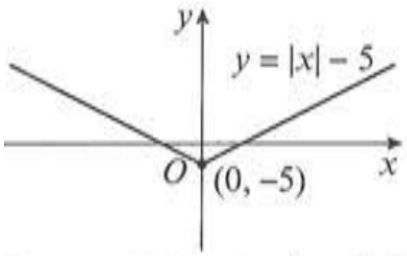
a. $\mathrm{f}(x) \geq 4$

b. $x = 11$

c. $4 < k \leq 10$

5.

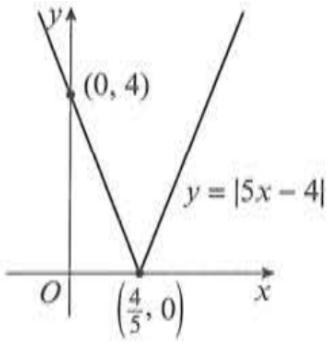
a.



b. For $x \geq 5$, $|x| - 5 = |x - 5|$. For $x < 5$, $|x| - 5 < |x - 5|$.
So $|x| - 5 \leq |x - 5|$ for all real values of x.

6.

a.



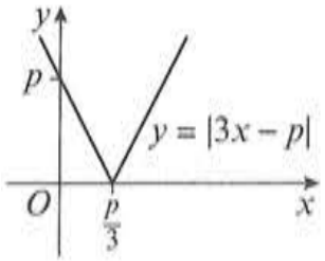
b. $\left\{x : x < \frac{1}{3}\right\} \cup \{x : x > 1\}$

c. $\left\{x : x < \frac{4}{5}\right\} \cup \left\{x : x > \frac{4}{5}\right\}$

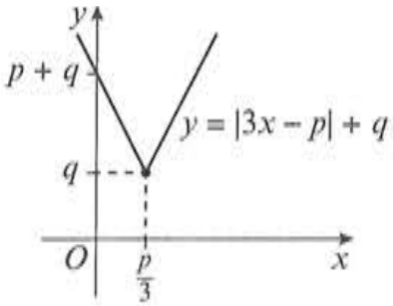
7.

a.

i.



ii.



b. $k = 2p$

Problem solving: Set A

Bronze

a. $\mathrm{f}^{-1}(x) = \sqrt[3]{\frac{3 - x}{2}}$

b. $\text{gf}(x) = \dfrac{10x^3 - 13}{3 - 2x^3}$

c. $x = \sqrt[3]{\dfrac{4}{3}}$

Silver

a. $\text{f}^{-1}(x) = \dfrac{\ln(x-2)}{2}$, $x \in \mathbb{R}$, $x \geq 2$

b. $\text{fg}(x) = (x+3)^2 + 2$, $\text{fg}(x) \geq 2$

c. $x = -11$ or $x = 5$

Gold

a. $\text{f}(x) = \dfrac{2x-3}{x-1}$, $x > 1$

b. $\text{f}(x) < 2$

c. $\text{f}^{-1}(x) = \dfrac{x-3}{x-2}$, $x < 2$

d. $x = \pm\dfrac{5\sqrt{6}}{9}$

Problem solving: Set B

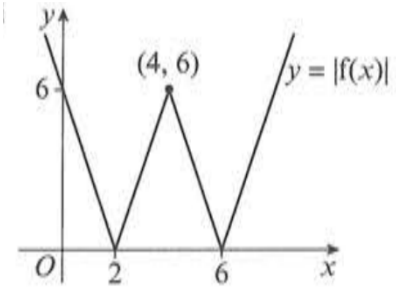
Bronze

a. 6

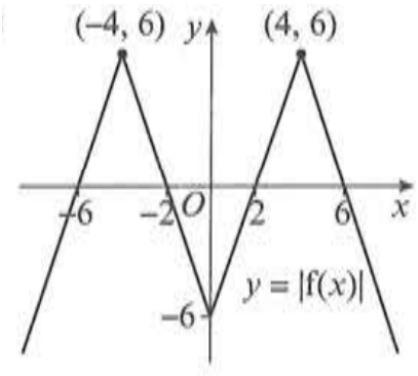
b. $\text{f}(x) \leq 6$

c.

i.



ii.



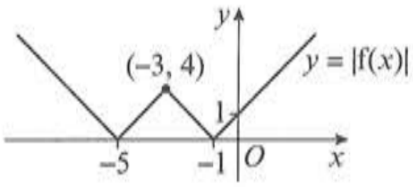
d. $x = \dfrac{14}{5}, \dfrac{34}{7}$

Silver

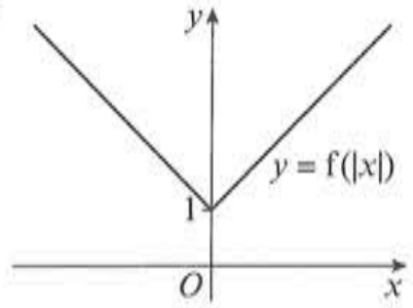
a. $\text{f}(x) \geq -4$

b.

i.



ii.



c. $\left\{x : -\frac{28}{3} \leq x \leq \frac{4}{5}\right\}$

d.

i. $k < -4$

ii. $k = -4$

iii. $k > -4$

Gold

a. $a = 6, b = 2$

b.

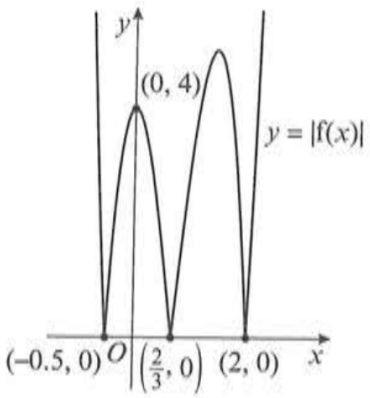
i. $\left\{p : -1 \leq p < -\frac{1}{2}\right\}$

ii. $\left\{-\frac{1}{2}\right\} \cup \{p : p < -1\} \cup \{p : p \geq 1\}$

iii. $\left\{p : -\frac{1}{2} < p < 1\right\}$

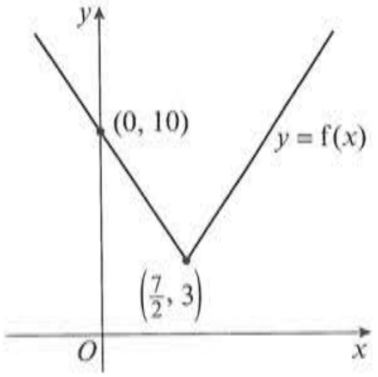
Selected exam question bank answers

Question 26 $\mathrm{f}(x) = (x-2)(3x-2)(2x+1)$



Question 32

a.



b. $\mathrm{f}(x) \geq 3$

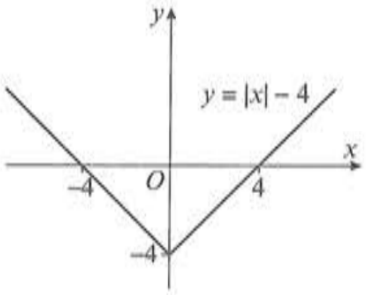
c. $\left\{x : x < \frac{8}{3}\right\} \cup \{x : x > 6\}$

Question 55

a. $x = -\frac{13}{3}, -3$

b.

i.



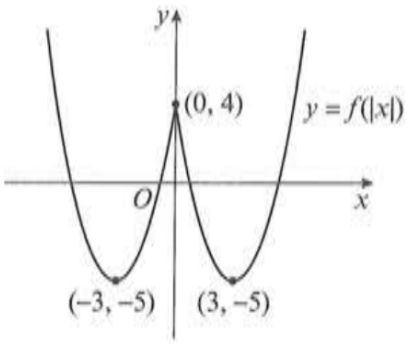
ii. For $|x| < 4$, $|x| - 4$ is negative and $|x - 4|$ is always non-negative, so $|x| - 4 < |x - 4|$
For $x \geq 4$, $|x| - 4 = |x - 4|$
For $x \leq -4$, $|x - 4| = 4 - x$ and $|x| - 4 = -4 - x$
So $|x| - 4 \leq |x - 4|$

Question 63

a.

i. $(6, -10)$

ii. $(-3, 5)$



b. $\mathrm{f}(x) = (x - 3)^2 - 5$

d. The function is not one-to-one.

Question 66

a. $\mathrm{f}(x) = \dfrac{2x-5}{x-2}$, $x > 2$

b. $\mathrm{f}(x) = 2 - \dfrac{1}{x-2}$, $x > 2$

c. T_1 is translation by $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$, T_2 is a reflection in the x-axis,

T_3 is translation by $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$

Question 79

a. $\mathrm{f}^{-1}(x) = \dfrac{4x+2}{x+3}$, $x \in \mathbb{R}$, $x \neq -3$

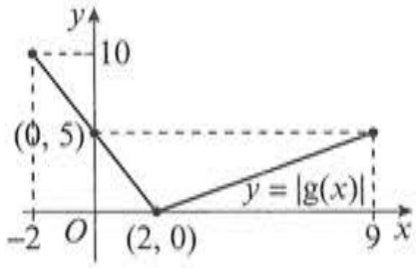
b. $-10 \leq \mathrm{g}(x) \leq 5$

c. -5

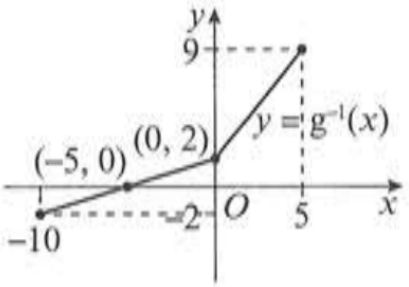
d. -13

e.

i.



ii.

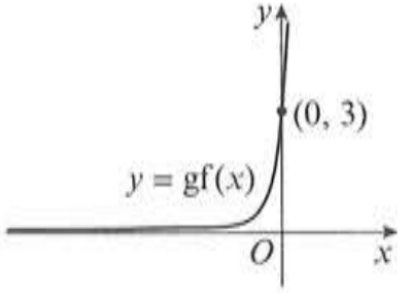


f. $-10 \leq x \leq 5$

Question 82

a. $\mathrm{gf}(x) = \mathrm{g}(3x + \ln 3) = e^{3x+\ln 3} = e^{3x} \times e^{\ln 3} = 3e^{3x}$

b.



c. $\mathrm{gf}(x) > 0$

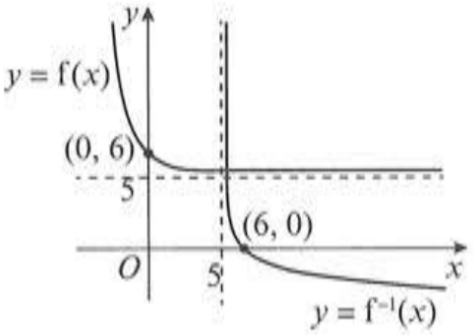
d. -0.270

Question 85

a. $\mathrm{f}(x) > 5$

b. $\mathrm{f}^{-1}(x) = \ln\left(\dfrac{1}{x-5}\right)$, $x \in \mathbb{R}$, $x > 5$

c.



d. Graphs meet on the line $y = x$, so at point of intersection

$$e^{-x} + 5 = x \Rightarrow \frac{1}{e^x} = x - 5, \text{ so } e^x(x-5) = 1$$